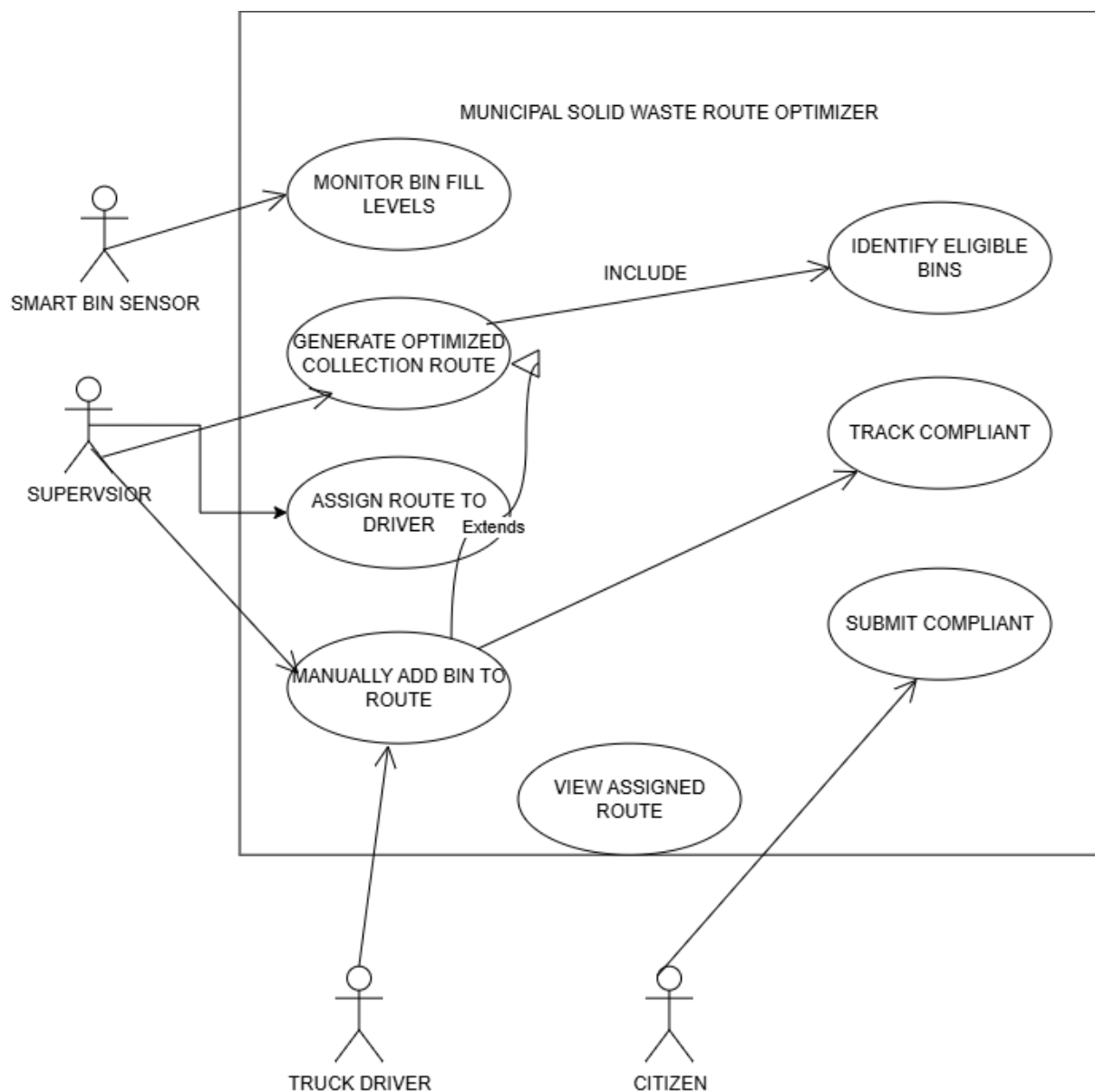


Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall generate optimized daily collection routes including only bins that have exceeded an 80% fill threshold.	High	Pass: Route excludes bins below 80% full unless manually tagged. Fail: An overflowing bin is omitted from the driver manifest.	Ensures trucks prioritize bins requiring collection.
FR-002	Functional	The system shall collect and display the fill-level data of registered smart trash bins.	High	Pass: The supervisor can view the latest fill percentage of every registered bin. Fail: A registered bin with available sensor data is missing from the dashboard.	Enables the supervisor to identify bins requiring collection.
FR-003	Functional	The system shall assign optimized collection routes to available garbage truck drivers.	High	Pass: Each generated route is assigned to an available truck driver and appears in the driver's route list. Fail: A generated route remains unassigned.	Ensures drivers receive actionable collection routes.
FR-004	Fucntional	The system shall allow the sanitation supervisor to manually add	Medium	Pass: A supervisor can select a bin and add it to an existing	Allows urgent or exceptional bins to be collected even when

		a bin to a collection route.		route, after which the bin appears in the driver's route list. Fail: The selected bin does not appear in the updated route.	they do not meet the automatic threshold.
FR-005	Functional	The system shall allow citizens to submit complaints for missed garbage pickups.	medium	Pass: A submitted complaint receives a unique complaint ID and is visible to the sanitation supervisor. Fail: A submitted complaint is not recorded or cannot be retrieved.	Helps the sanitation department track and resolve missed pickups.
NFR-001	Nonfunctional	Route recalculation for a fleet of 50 trucks across 10,000 bins shall execute in under 45 seconds.	High	Pass: Benchmarking confirms route recalculation completes within 45 seconds under the specified workload. Fail: Any benchmark exceeds 45 seconds.	Ensures the system can respond quickly for a large city-scale fleet.
NFR-002	Nonfunctional	The system shall restrict route management functions to authenticated	High	Pass: Only authenticated users with supervisor privileges can generate,	Prevents unauthorized modification of garbage collection operations.

		and authorized sanitation supervisors.		modify, or assign collection routes. Fail: An unauthorized user can access any route management function.	
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3.USER CASE FLOW DOCUMENT

Primary Actor

Sanitation Supervisor

Preconditions

1. The sanitation supervisor is authenticated.
2. Smart bins have provided their latest fill-level data.
3. Truck and driver information is available in the system.

Postconditions

1. An optimized collection route is generated.
2. Only eligible bins are included in the route unless manually tagged.
3. The generated route is available for assignment to a truck driver.

Main Success Scenario

1. Sanitation Supervisor logs into the municipal sanitation portal.
2. System displays the current fill levels of registered smart bins.
3. Supervisor selects **Generate Optimized Collection Route**.
4. System identifies bins whose fill level exceeds 80%.
5. System retrieves the available trucks and drivers.
6. System calculates an optimized collection route for the eligible bins.
7. System displays the generated route and the bins included in it.
8. Supervisor reviews the generated route.
9. System saves the route.
10. Use case ends successfully.

Alternate Flow — No Eligible Bins

4a. No bin exceeds the 80% threshold

1. System determines that no registered bin has exceeded 80% fill level.
2. System displays a message stating that no collection is currently required.
3. Supervisor may choose to manually add a bin if required.
4. Use case ends.

Alternate Flow — Manual Addition

You can also add this if you have space:

7a. Supervisor identifies an urgent bin

1. Supervisor selects **Manually Add Bin**.
2. System displays available bins.
3. Supervisor selects the required bin.
4. System adds the bin to the route.
5. System recalculates the route.
6. Updated route is displayed.

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SECTION :H

SCENARIO: 25.Municipal Solid Waste Route Optimizer